Zer0pa / ZPE-Neuro

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Watch 0

ZPE-Neuro V0.0: DETERMINISTIC NEURAL SIGNAL CODEC: Extracellular Spikes | Neuropixels | DANDI-Anchored | IBL Validated | Spike Train Transport

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Zer0pa-Architect-Prime docs: collapse proof surface and sync main ✓ 9560e86 · 1 hour ago

.github	docs: collapse proof surface and sync main	1 hour ago
.gpd	feat: record lane 1 authority phases throug...	9 hours ago
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tests	chore: align release surface and documen...	3 hours ago
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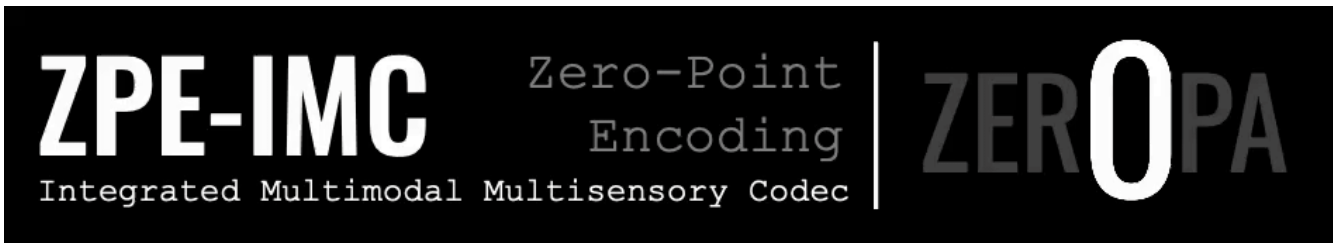
ZPE-Neuro



license SAL v6.0 | python 3.11 | authority 2026-03-21 repo snapshot | release private staged | lane extracellular recording

quick verify audit path | architecture runtime map | limits honest reading | docs canonical registry

What This Is	Current Authority	Runtime Proof	Quickstart & License
Proof Corpus	Open Risks	Go Next	Docs Registry

WHAT THIS IS

What This Is

ZPE-Neuro is the inner repository for the Wave-1 neural signal package, the current curated proof corpus, and the release-surface documentation for the Neuro workstream. It is a private staged repo, not a tagged public release.

The current scope lock is a narrower extracellular lane. The strongest honest repo claim is not broad neural generality; it is a truthful Pythc package surface plus a bounded evidence surface anchored on DANDI `000034` , a counted IBL second-target `PASS` under bounded refinement, explicit AJILE12 out-of-family handling, and unresolved blind-clone/commercialization gates.

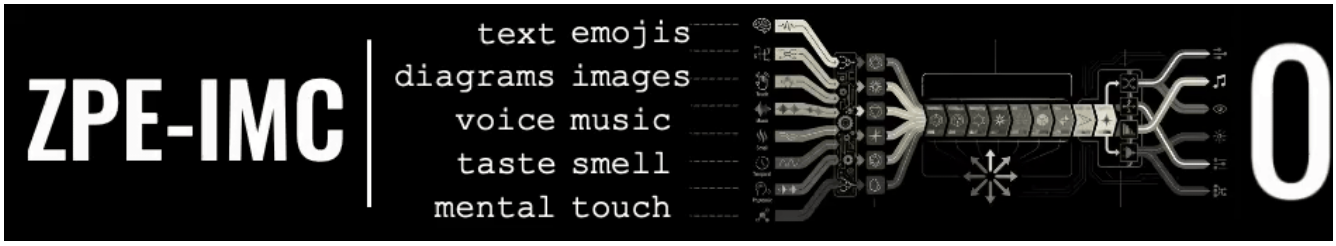
CURRENT AUTHORITY

Current Authority

Use this table as the single canonical authority block for the repo front door.

Field	Current truth	Authority
Authority snapshot date	`2026-03-21`	proofs/manifests/CURRENT_AUTHORITY_PACKET.md
Repository / acquisition surface	Canonical GitHub repo: `https://github.com/Zer0pa/ZPE-Neuro` for authorized readers. Clone surface for authorized readers: `git clone https://github.com/Zer0pa/ZPE-Neuro.git` .	`AUDITOR_PLAYBOOK.md` , `docs/ARCHITECTURE.md`
Repo posture	`PRIVATE_STAGED` . Package, install, docs, and proof surfaces are aligned for the current repo state, but this is not a public release and not a clean-clone-closed authority packet.	`RELEASING.md` , `PUBLIC_AUDIT_LIMITS.md`
Top unresolved gate	A fresh clean-clone replay of the authority packet remains the top unresolved acceptance gate.	`RELEASING.md`

Field	Current truth	Authority
Gate status	`OPEN` for blind-clone and public-release gates. `PASS` for the current clean packaged baseline and the tracked release-alignment gate slice.	`proofs/selected_artifacts/2026-03-21_zpe_neuro_release_alignment/`
Current lane / scope lock	Lane 1 is a narrow extracellular recording lane. AJILE12 is explicitly `OUT_OF_FAMILY` for the current codec; broader human/intracranial coverage is not claimed here.	`proofs/selected_artifacts/2026-03-21_zpe_neuro_ibl_refinement/public_corpus_summary.j`
Primary positive public anchor	DANDI `000034` remains the strongest positive public waveform anchor in the current repo surface.	`proofs/selected_artifacts/2026-03-21_zpe_neuro_ibl_refinement/public_corpus_summary.j`
Counted breadth verdict	`PASS` in the current bounded local evidence packet after the March 21 IBL refinement. This does not upgrade blind-clone status or broader release claims.	`proofs/selected_artifacts/2026-03-21_zpe_neuro_ibl_refinement/public_corpus_summary.j`
Family-boundary decision	`OUT_OF_FAMILY` for AJILE12 in the current lane.	`proofs/selected_artifacts/2026-03-21_zpe_neuro_ibl_refinement/public_corpus_summary.j`
Packaged clean baseline	`PASS` for the packaged `.`, `.[gate]`, `.[public]`, and `.[proof]` surfaces that are actually declared for clean install/import. IBL chunked-waveform tooling and Allen parity remain operator-only.	`proofs/selected_artifacts/2026-03-21_zpe_neuro_release_alignment/verification_summary` `pyproject.toml`, `docs/ARCHITECTURE.md`
Blind-clone verification status	`OPEN`	`PUBLIC_AUDIT_LIMITS.md`, `RELEASING.md`
Release status	`NO_PUBLIC_RELEASE`. The current repo surface is documented and internally coherent, but no public-release verdict is claimed.	`RELEASING.md`
Commercialization status	`OPEN`. Allen parity and commercialization closure remain unresolved.	`docs/LEGAL_BOUNDARIES.md`, `RELEASING.md`
Primary authority artifacts	`proofs/manifests/CURRENT_AUTHORITY_PACKET.md`, `proofs/selected_artifacts/2026-03-21_zpe_neuro_release_alignment/`, `proofs/selected_artifacts/2026-03-21_zpe_neuro_ibl_refinement/`	`proofs/README.md`
Audit route	Start with the short replay path, then read the limits note before widening any claim.	`AUDITOR_PLAYBOOK.md`, `PUBLIC_AUDIT_LIMITS.md`



RUNTIME PROOF (WAVE-1)

Runtime Proof And Package Truth

Surface	Current state	Why it matters
Core package build/import	`PASS`	The repo now ships a truthful Python package surface rather than relying on undeclared runtime assumptions.
Repo-local tests	`PASS` on the current shipped unit slice	The staged code surface has a small but real regression check that ships with the repo.

Surface	Current state	Why it matters
Synthetic gate baseline	`PASS` for sequential Gate C and Gate D replay in the current clean packaged gate stack	This is the strongest current shipped replay baseline in the repo.
Public replay import surface	`PASS` for the declared `.[proof]` import stack	This keeps the docs honest about what the packaged replay stack actually installs cleanly.
IBL and Allen operator paths	`OPERATOR_ONLY`	Those paths currently require manual dependency/toolchain work around `ONE-api`, `ibl-neuropixel`, `llvmlite` / `numba`, or `allensdk` conflicts and are intentionally not shipped as clean extras.
Release automation	`PASS` for static verification coverage	The repo now has a verification workflow that checks package/build/install truth without implying a live publish pipeline or automated publish step.

QUICKSTART AND LICENSE

Quickstart And License

Acquire from the private GitHub repo if you have authorized access, then verify the shipped package surface from a checkout.

```
git clone https://github.com/Zer0pa/ZPE-Neuro.git
cd ZPE-Neuro

python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -e .
python -c "import zpe_neuro"
```

Repo-local test slice:

```
python -m pip install -e '.[dev]'
python -m pytest tests
```

Repo-local synthetic gate slice:

```
python -m pip install -e '.[gate,dev]'
python tools/run_gate_c.py --artifact-root artifacts/manual_gate_c --seed 20260220
python tools/run_gate_d.py --artifact-root artifacts/manual_gate_d --replay-seeds 20260220,20260221,20260222,20260223
```

Repo-local clean public replay import surface:

```
python -m pip install -e '.[proof]'
```

The `tools/` runners are repo-local scripts, not installed console entry points. `LICENSE` is the legal source of truth. Read `docs/LEGAL_BOUNDARIES.md` before turning any repo result into a wider legal or commercial claim.

PROOF CORPUS

Proof Corpus

Packet	Class	How to read it
`proofs/manifests/CURRENT_AUTHORITY_PACKET.md`	`CURRENT`	The routing manifest for what is current, what is historical, and which packet owns each claim layer.
`proofs/selected_artifacts/2026-03-21_zpe_neuro_release_alignment/`	`CURRENT`	Tracked summaries for the March 21 release-alignment technical pass.
`proofs/selected_artifacts/2026-03-21_zpe_neuro_ibl_refinement/`	`CURRENT`	Current bounded local extracellular breadth packet, including the counted IBL second-target `PASS` .
`CHANGELOG.md` and `runbooks/README.md`	`SUPPORTING`	Chronology, receipts, and operational history. Use them to understand how the current repo state was reached, not as current proof authority.

OPEN RISKS

Open Risks

Surface	Class	Current truth
Blind-clone authority pack	`OPEN`	The repo does not yet prove a fresh blind-clone authority replay.
Public release	`OPEN`	No tagged/public release readiness is claimed from this repo state.
IBL / Allen operator paths	`OPERATOR_ONLY`	These remain outside the clean packaged release surface because the upstream dependency chains are not currently truthful for a clean install.
Commercialization closure	`OPEN`	Allen parity and commercialization-safe closure remain unresolved.
Broader neural scope	`PARKED_BY_SCOPE`	Broader human/intracranial or second-mode coverage is outside the current lane.
Historical path residue	`KNOWN_RESIDUE`	Some tracked runtime artifacts still contain machine-absolute paths inside captured traces. They are evidence lineage, not current filesystem instructions.

WHERE TO GO

Where To Go

Need	Start here
Shortest honest audit path	`AUDITOR_PLAYBOOK.md`
Current authority routing	`proofs/manifests/CURRENT_AUTHORITY_PACKET.md`
Architecture and package boundaries	`docs/ARCHITECTURE.md`

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 1

Zer0pa-Architect-Prime

Languages

- Python 74.3%
- xBase 25.7%